A 64-Channel, Low-Power End-to-End Transformer-Based BCI SoC for Efficient Motion Intention Decoding

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Abstract

Real-time brain-computer interfaces (BCIs) demand fully on-chip neural decoding with tight power, area, and latency budgets. This work presents the first fully-integrated transformer-based BCI SoC that performs energy-efficient motion intention decoding from ECoG signals. The chip integrates a 64-channel analog front-end (AFE) and a distilled, light-weight transformer-based decoder featuring customized 7T-SRAM computing-in-memory (CIM) macros with hybrid sign-magnitude and two's-complement (SM-2C) computation. Fabricated in 65nm, the decoder achieves a 67.8% accuracy on a 6-class upper-limb motion-intention dataset, consuming 47µW in 4.7mm², while the complete SoC occupies 10.1mm².

Introduction

Restoring motor function in patients with tetraplegia caused by stroke or spinal cord injury remains a key goal in modern neuro-engineering. Prior work has demonstrated restoration of lower-limb motion by decoding neural signals on an external computer and stimulating the spinal cord [1]; however, such backpack-mounted or externally tethered systems incur large latency (>100ms) and limit practical, unobtrusive use. Achieving portable, real-time operation therefore requires fully on-chip decoding. Recent neural-decoding SoCs [2-6] have made progress toward this goal, but most rely on off-chip memory or are constrained by limited on-chip storage and compute resources, restricting the adoption of advanced models such as transformers for robust motion intention decoding.

To the authors' knowledge, this work demonstrates the first fully-integrated transformer-based BCI SoC for motion intention decoding with compact area, low power, and low latency. As shown in Fig. 1, key features include: (1) an end-to-end system integrating a 64-channel low-noise, low-power AFE, a wavelet-based feature extractor, and an on-chip backend decoder; (2) a lightweight knowledge-distilled transformer with frame-by-frame processing, approaching the performance of a software-based ~100-M-parameter model while significantly reducing feature-memory overhead; and (3) custom 7T-SRAM CIM macros with hybrid SM-2C computation, enabling low-power classification with 0.224-ms compute latency and a 40% reduction in decoder power.

Hardware Implementation

Fig. 2 overviews the proposed motion-decoding BCI. A 64-channel AFE amplifies and digitizes ECoG signals, which are down-sampled to 500 Hz and passed through a wavelet module to generate 10-frame, 320-D features. These features are fed to a lightweight (~30K parameters) BrainDistill [7] transformer. Knowledge distillation reduces the parameter count by ~2800× while achieving performance comparable to a software-based ~100-M-parameter transformer model. The system operates in a frame-by-frame processing flow rather than batching entire 10-frame windows, reducing ~90% of memory access between the wavelet module and the decoder. All operations are orchestrated by an on-chip finite-state machine (FSM) (Fig. 2, bottom), enabling fully on-chip processing without off-chip memory access.

The 64-channel AFE (Fig. 3) builds upon [8] and achieves a 30% area reduction while maintaining signal fidelity for neural decoding. The AFE is partitioned into four 16-channel modules, where a multiplexer routes signals from the LNAs to a shared Gm-C filter and a 10-bit SAR ADC operating at 64kS/s (4kS/s/channel). Area reduction is achieved through architectural optimization of the AC-coupled LNA and shared backend resources.

Fig. 4 presents the block diagram of the proposed transformer-based decoder. It integrates two SM-CIM arrays for input projection and four 2C-CIM arrays for the two linear-attention layers. Two weight SRAMs store parameters for the remaining layers, including QKV computation, output projection, and layer normalization. Overall, 66% of parameters and 72% of MACs are implemented using the custom CIM arrays, resulting in a 40% reduction in decoder power. Two feature SRAMs store intermediate activations and are reused across the two linear attention layers, reducing feature-memory usage by 4.4× (from 61.44kb to 14.08kb). Together with the remaining digital modules, the decoder forms a compact, fully on-chip end-to-end BrainDistill network for motion intention classification.

Inspired by [9], the proposed 7T-SRAM CIM macro supports isolated read/multiply (V_{DDH}) and retention/write (V_{DDL}) operations (Fig. 4). Monte-Carlo simulation shows reliable write operation for V_{DDH}>0.7V. In the SM-CIM macro, each 8-bit weight is decomposed into sign and magnitude bits and stored in eight vertically stacked 7T-CIM cells, enabling bit-serial 8-bit SM multiplication. In the 2C-CIM macro, an 8-bit 2C input is distributed through a MUX to eight parallel 7T-CIM cells, whose outputs are accumulated by a shared 2's-complement accumulator.

To further improve energy efficiency, hybrid SM-2C computation is adopted based on weight and activation statistics (Fig. 6). Because SM representation reduces bit toggling for small values, SM-CIM is used for input projection, where activations cluster near zero. In contrast, FC1/FC2 leverage 2C-CIM to support broader activation ranges and avoid format-conversion overhead, reducing the overall toggle rate by 30%. In SM-CIM array, the input activation is broadcasted across 32 WLs in bit-serial manner. The partial sums from different WLs are subtracted ($AP_{0-31}-AN_{0-31}$) and accumulated in parallel. In the 2C-CIM arrays, partial sums are accumulated within each macro and then combined using an adder tree to produce the final outputs.

Measurement Results

Fig. 7 shows the measurement setup. The SoC occupies 10.1mm² (Fig. 8). The AFE dissipates 3.56µW/channel, providing a programmable gain from 33 to 66dB, an SNDR of 45dB (Fig. 2, bottom) and an input impedance of 10.6MΩ at 1kHz. The decoder consumes 47µW, with 19% dynamic and 81% leakage power (Fig. 9). Thanks to the CIM architecture, dynamic/leakage power is reduced by 73%/18%, respectively. The chip achieves software-level performance on a 6-class upper-limb dataset, recorded with WIMAGINE device [10]. Fig. 10 shows the predicted motion probabilities. Table I benchmarks the decoder with prior art, demonstrating the lowest power and state-of-the-art accuracy, while supporting a larger model size and more classes. This is the first BCI SoC for upper-limb movement decoding, whereas prior works primarily focus on finger-level or simple gesture decoding.

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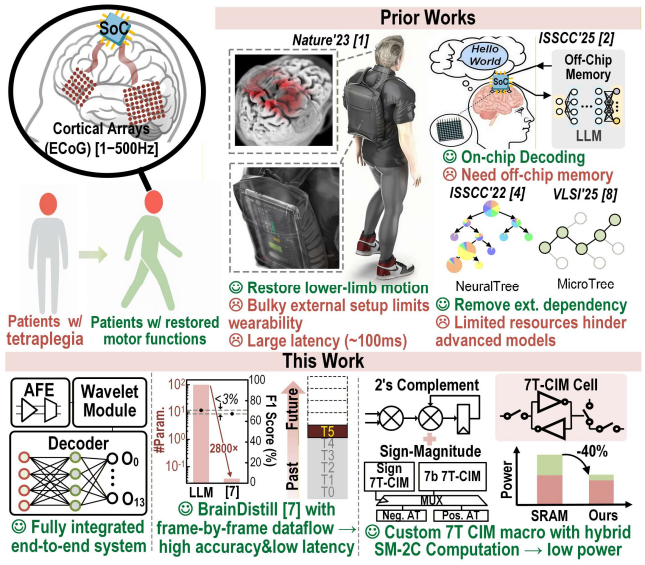


Fig. 1. Prior work (top) and contributions of this work to address shortcomings of prior art (bottom).

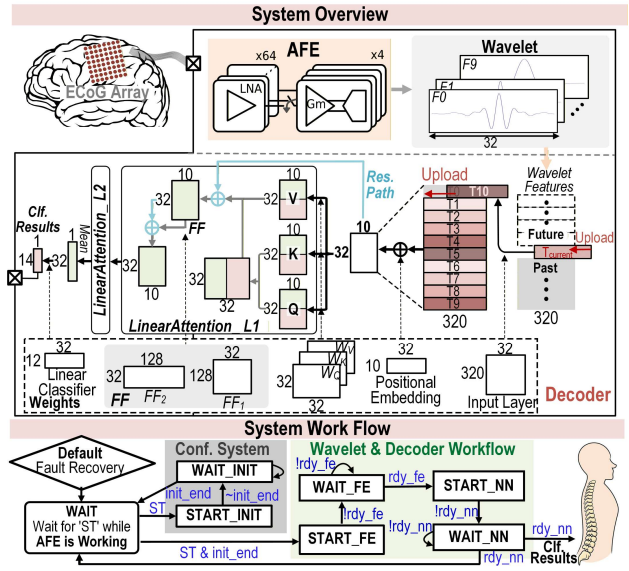


Fig. 2. System overview with decoder data flow diagram (top) and system control state machine (bottom).

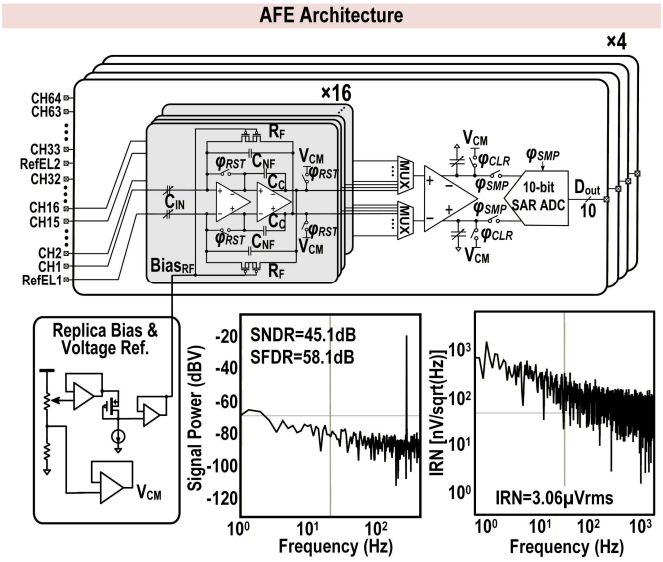


Fig. 3. 64-channel AFE architecture (top) and measurements (SNDR: bottom-left and input-referred noise: bottom-right).

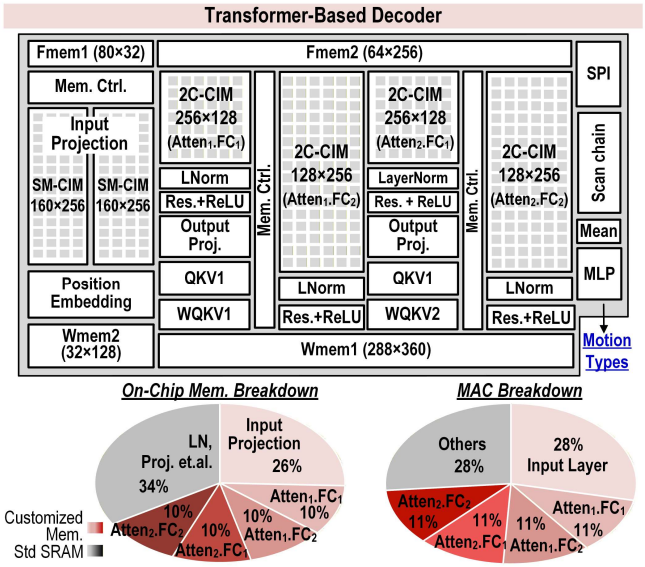


Fig. 4. Architecture of the proposed transformer implementation for motion-intention decoding.

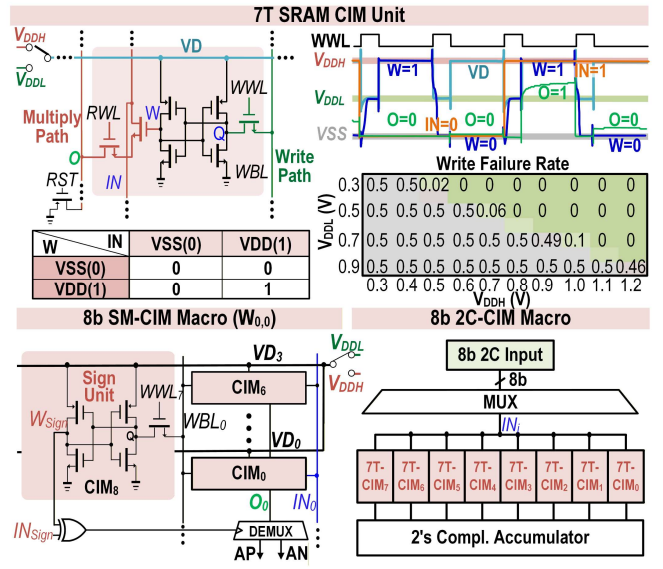


Fig. 5. Proposed 7T-SRAM CIM implementation (top) and the SM-CIM and 2C-CIM macro (bottom).

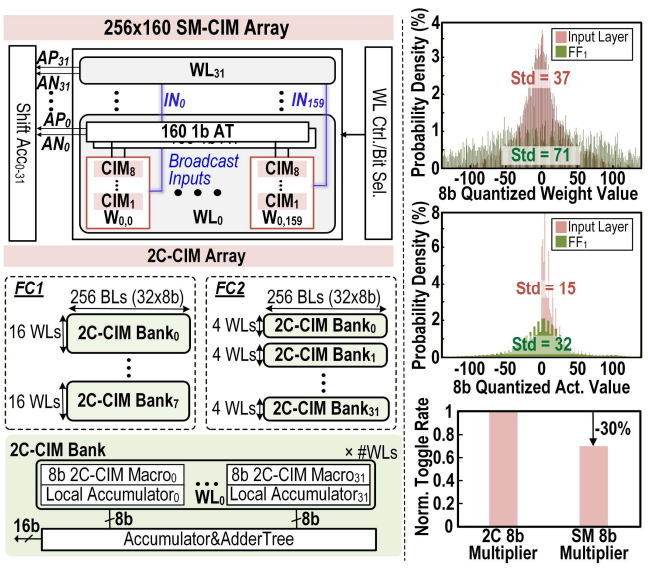


Fig. 6. Proposed hybrid SM-2C CIM architecture with reduced toggle activity.

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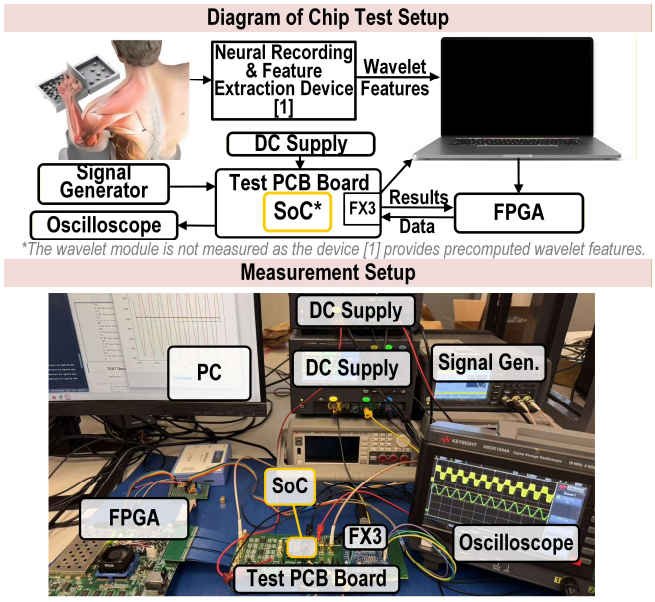
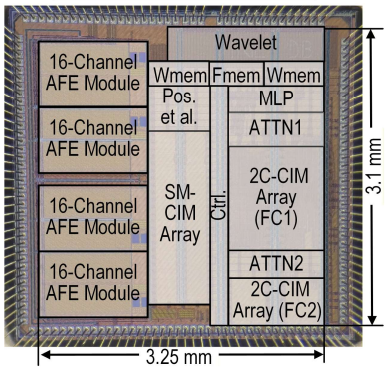


Fig. 7. Chip testing setup and measurement platform.

AFE Specifications	
Technology	65nm
Supply Voltage (V)	1.2
# of Channels	64
Area/Ch. (mm²)	0.038
Power/Ch. (µW)	3.56
IRN (µVrms)	3.06



Decoder Specifications		
Technology		65nm
Supply Voltage (V)		0.5/1.2
Frequency (MHz)		10.24
Core Area (mm²)		1.87×2.49
On-chip SRAM (kB)	Foundry SRAM	9.47
	CIM-SRAM	26
Supported Precision		INT8
Comp. Latency (ms)		**0.224**
Power (µW)		**47**
Energy Consumption (nJ/Classification)		**10.53**
Energy Efficiency (TOPS/W)		**68.6**
Accuracy (%)		Upper-limb: **67.8** Finger: **77**
F1-Score (%)		Upper-limb: 68 Finger: 75.7

Fig. 8. Die photo and chip summary.

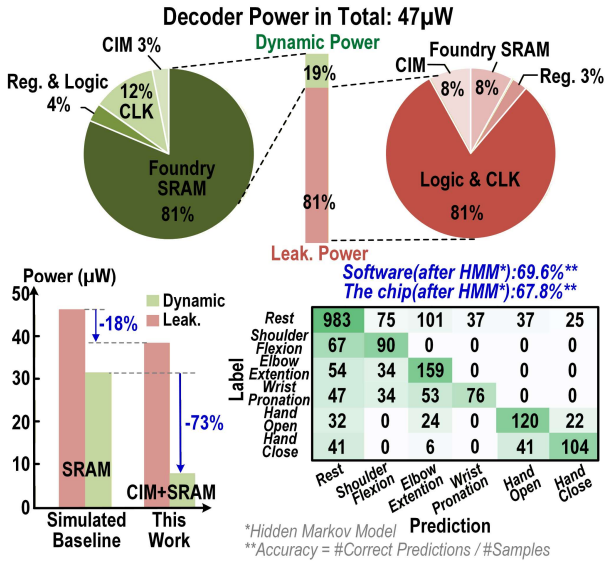


Fig. 9. Decoder power breakdown and decoding confusion matrix.

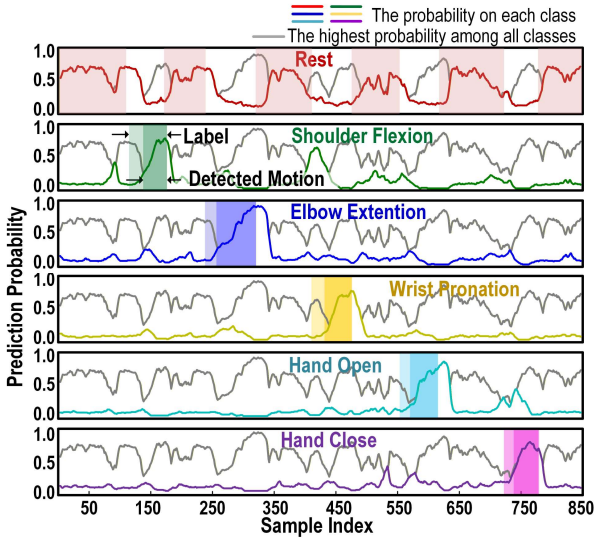


Fig. 10. Upper limb example decoding results: the BrainDistill implementation successfully decodes movement intention vs resting state (top) with low latency.

TABLE I. Comparison with state-of-the-art BCI SoCs.

		This Work	Nature'23 [1][a]	ISSCC'24 [3]	ISSCC'22 [4]	TBioCAS [5]
	Technology	65nm	-	65nm	65nm	180nm
Decoder	Supply (V)	0.5/1.2	-	1.2	1.2	0.625
	# Channels	64	64	16	256	93
	Clock Freq. (MHz)	10.24	-	-	0.128	31.5
	Decoding Model	**Transformer**	HMM+ Regression Models	DFT+CNN	NN+ Decision Tree	SBP+ SSKF
	Model Size (KB)	**35**	-	15.8	2.93	<1
	On-Chip Mem. (KB)	35.47	-	-	2.93	-
	Decoding Task	Upper-limb Motion & Finger Movements	Lower-limb Motion	Motor Imagery	Finger Movements	Finger Movements
	Input Signals	ECoG	ECoG	EEG	ECoG	Intracortical
	Supported # Classes	14	7	2	6	Continuous
	Accuracy(%)	**67.8**(Upper-limb Movements)/ **77**(Finger Movements)	74	79.2-94.8	73.3	96(2D)[b]
	Power (µW)	**47**	-	52.8	158	587.8
	Power/Ch. (µW)	**0.73**	-	3.3	2.47[c]	6.3
	Latency (ms)	**0.224**	-	-	<1000	50
	SoC	**Yes**	No	Yes	Yes	Yes
AFE	Area/Ch. (mm²)	0.038	-	0.16	0.004	0.615
	Power/Ch. (µW)	3.56	-	8	1.51	129
	IRN (µVrms)	3.06	-	-	3.2	2.4

[a] Running with Mathworks on PC [b] Different dataset [c] Selective 64-channel processing

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